

Project Proposal

Academic Year 2024

Candidate Details

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Supervisor/Advisor Details

	Supervisor 1 (IT Related)	Supervisor 2 (Optional)
Name	Mr. M.P. Sarath Wijesinghe	
Designation	Assistant Director - Information Technology	

Workplace address	Sri Lanka Export Development Board, No.42, Nawam Mawatha, Colombo 2.	
Academic/ professional qualifications and memberships	Master of Information Technology, University of Kelaniya Post Graduate Diploma in Information Technology, University of Kelaniya. Degree in Statistics and Computer Science, University of Sabaragamuwa. Certificate in PHP and MySQL, Sri Lanka Institute of Information Technology (SLIIT) Certificate in Web Development, Sri Lanka Institute of Information Technology (SLIIT) Certificate of Creating Quality E-commerce Content conducted by SME Trade Academy, The International Trade Centre (ITC), Palais des Nations, 1211 Geneva 10, Switzerland. Certificate Verification Link: https://learning.intracen.org/verify/?q=s8YMTJ75uD	
Work experience	Assistant Director – Information Technology Sri Lanka Export Development Board (EDB) Duration: 15+ years E-Commerce and Software Development: Over 12 years of expertise in managing and developing e-commerce platforms and software solutions to enhance trade facilitation and operational efficiency.	

	• <i>Project Supervision: 6 years of experience supervising projects at the tertiary level, including guiding university students and professionals in software development and IT-related initiatives.</i> • <i>IT Project Implementation: Proven track record of successfully implementing IT projects for 15 years, contributing to the digital transformation of organizational processes and service delivery.</i>	
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1. Title of Project

Student Management System For an Educational Institute.

2. Client Information (If applicable)

- **Name:**
- **Address:**
- **Contact Information:**

3. Introduction

This system will be implemented where all the tasks are done manually. It provides student Registration, Maintaining student profiles, enrollment details, tracking their attendance, financial Operations, grading system, teacher's profiles, and essential learning materials. It enables efficient administration and allows students to complete their educational process easily.

Furthermore, the website provides more details and as well a user-friendly interface.

Today we live in a digital world and technology can make our work easier.

A student Management Tool is an important tool for a tuition class. It saves time, improves Efficiency improves communication between users and most importantly students can learn remotely. I believe a well-implemented student management system leads to better organization, enhanced learning outcomes and a more efficient educational system.

4. Motivation

Many educational institutes still rely on manual processes or obsolete software for managing their administrative work. It can be time-consuming and inefficient. With a fully digitized system, it ensures smoother communication between administrators, teachers, students and. It also enables features like fee management and attendance tracking. Hence it saves a lot of time

and reduces administrative burden. Having a digital system is an advantage for all stakeholders

-It simplifies complex administrative processes for administrators.

-Easy to manage attendance, grades and better connection between students.

-Can easily view their schedules and access to study material for students.

-Reduces paper-based process and operational cost to institute and can attract more students

With the modern technology-driven reputation.

5. Objectives

- Replace administrative processes with an efficient automated system.
 - Save time and eliminate errors by streamlining manual procedures such as student registration attendance monitoring, and grade management.
- Maintain and centralize student profiles, enrollment details, teacher's profiles, financial details and administrative records.
 - Provide a single platform for securely storing and accessing all student, teacher and administration records.
- Improve accessibility
 - Enable users to access the system from anywhere anytime with internet connectivity.
- Improve efficiency
 - This system simplifies tasks like student fee payments, report generation, essential learning material sharing and reducing administrative burden.
- Facilitate Learning Material Management
 - Allow teachers to upload essential learning materials and student can easily access their relevant learning materials anytime.
- Enable important notifications
 - Improve communication by automating notifications for important events like as exam dates, exam results, assignment uploading and also due payments as well.

6. Scope

In scope

- Automate the Management of Student and Teacher Data
 - To increase data accessibility and accuracy, centralize procedures such as teacher profile enrollment, profile management, and student registration.
- Monitoring Attendance
 - The system tracks and reports students' attendance.
- Financial Operations
 - Managing student fees, dues and other financial operations.
- Student grading details
 - Showing student's grades and their progress
- Essential learning materials and teacher profiles
 - Maintain teacher profiles and upload relevant study materials for students.

Out of scope

- During the initial phase, the system will not integrate with external platforms such as finance applications.
- This system requires an internet connection and will not support offline operations

7. Methodology

This project will follow the waterfall approach. A sequential development approach where each phase is completed before moving to the next step. It is a clearly organized process with clear user requirements and ensures quality end product.

- Requirement Gathering - Gather data from the institute owner and administrative members through Interviews

- System Design - Design the system's architecture, database schema, and user interfaces for the collected requirements
- Development - Implement the modules using PHP, JQUERY and BOOTSTRAP 5.
- Testing - Perform unit and integration testing to recognize system bugs and ensure System functionality and stability.
- Deployment - Deploy on a local server for demonstration purposes and prepare final documentation for handover.
- Documentation - Create detailed user and technical documentation, including guidelines for system setup, maintenance, and future updates.

8. Requirements

• 8.1. Functional Requirements

- Student registration module
 - When a new student registers administrators will add new student records with the required fields.
 - Generate a unique student ID when registering a student.
 - Allow uploads of relevant documents during registration.
 - Generate a conformation message after a successful registration
- Grading and performance module
 - Teachers shall be able to input grades of exams and assignments.
 - Compare previous grades with current grades and analyze progress.
 - Generate individual grade reports.

- Fee management module
 - Administrators shall be able to access relevant fee amounts according to different subjects.
 - Generate an online payment receipt after a successful payment.
 - Generate a notification after a successful payment to students.
 - Send reminders about upcoming payments and due payments.
- Learning Materials Management Module
 - Allow teachers to upload learning materials according to their subjects.
 - Allow students to download materials easily.
 - Notify students when new material is uploaded.
- **8.1. Non Functional Requirements**
 - Performance-System should be able to handle a fair amount of users.
 - User-Friendliness-It should be easy to use for every kind of user.
 - Availability and Reliability-System should be available at any time and ensure minimal downtime
 - Security-ensuring that only authorized users can access the system with a username and password

9. Deliverables

- Fully functional student management system.
- User Manual.
- Project documentation.
- Training materials.
- Access Credentials.
- Source code
- Project handover report.

10. Feasibility Study

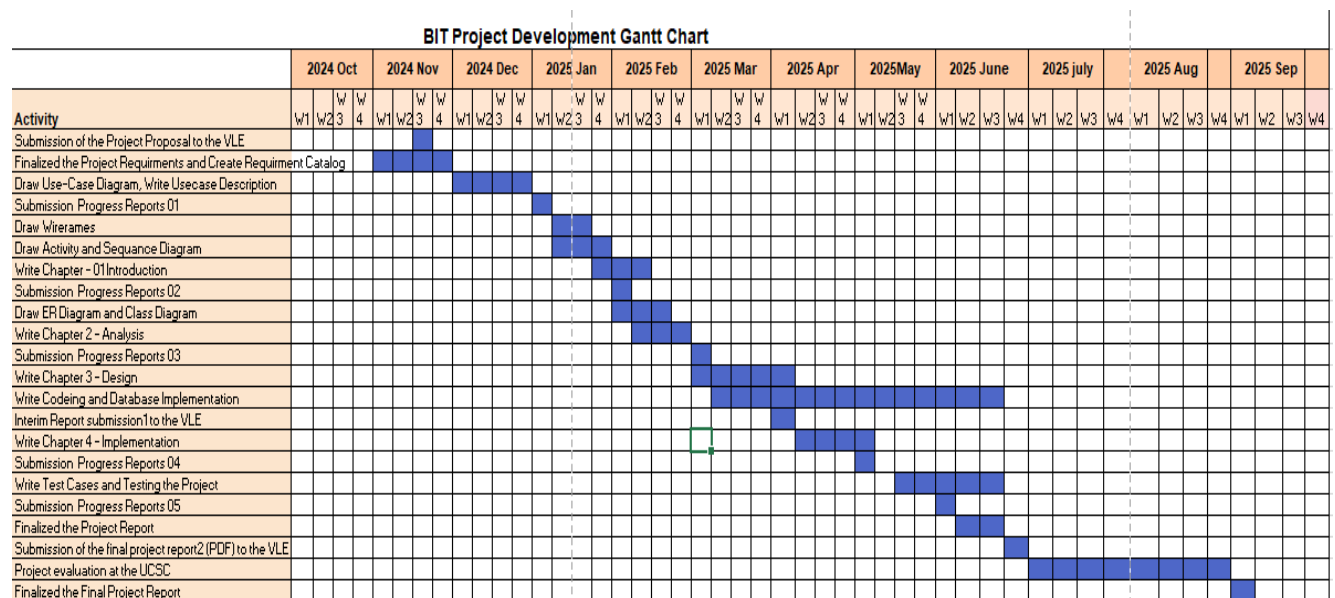
Technical Feasibility -This system will use used most suitable programming languages and tools for the backend, JQUERY for the front end and BOOTSTRAP 5 with H TML 5 used for a responsive and user-friendly interface. And these are widely available and well-supported.

Financial Feasibility -The project is financially feasible because it has the potential for cost reductions, efficient resource usage and long-term benefits that cover the initial cost. Well, planned and cost-effective tools can help stay within the institution's budget.

Operational Feasibility-This system is operationally feasible because of its user-friendliness, interoperability and scalability. proper training and assistance will result in a smooth transition and long-term success.

11. Project Plan

Gantt Chart: Include a visual representation of the project timeline and key



12. Resource Requirements

- Quad-core or higher processor
- Minimum 8Gb ram

- Minimum 500Gb hard disk
- Windows 8 OS or higher
- Reliable internet connection
- MySQL database management system
- A web server
- Testing tools

13. Evaluation Criteria

- User Feedback and Satisfaction surveys - It focus on collecting opinions and experiences from users. It mainly considers ease of use, effectiveness and performance of the system. It also helps to identify weaknesses and identify areas for improvement in the system.
- Usability Testing Metrics – This helps to recognize system issues and improve the overall user experience. This can be done by conducting unit testing, integrated testing and system testing.
- Functional Performance Analysis -Evaluate how effectively the system performs under normal and varying conditions. Mainly consider accuracy, efficiency and response time. This ensures the system meets functional requirements and performs ideally.
- Independent Reviews and Testing - involves third-party assessments to ensure fair feedback about the software. It mainly focuses on compliance and reliability. these reviews helps to identify and validate software quality and performance.

14. References

- PHP, jQuery, Bootstrap 5, and MySQL official documentation
- Best practices for building web applications
- Existing student management systems